

PHP and Working Procedure

PHP (Hypertext Preprocessor) is a general-purpose scripting language that is mainly used for server-side scripting.

When a user requests a webpage that contains PHP code (like index.php), the request goes to the web server. The server sees the .php file and hands it to the PHP parser. The parser executes all the PHP code on the server side; this can include tasks such as connecting to a database, verifying login credentials, or calculating a result. The PHP script then generates pure HTML output (or other content like JSON or XML), which is sent to the user's browser. The browser never sees the PHP code, only the final output. This is why PHP is called a server-side scripting language, and pages created this way are considered dynamic, because the content can change based on input, time, or data.

In short, PHP runs on the server, generates HTML output, and sends that to the browser. It makes web pages interactive and dynamic.

Practical & Useful PHP Code Example:

1.Simple Login Form

index.html (front-end)

```
<!DOCTYPE html>

<html>

<head>
    <title>Login Page</title>
</head>
<body>
    <form action="login.php" method="POST">
        <label>Username:</label><br>
        <input type="text" name="username" required><br><br>
        <label>Password:</label><br>
        <input type="password" name="password" required><br><br>
        <input type="submit" value="Login">
    </form>
</body>
</html>
```

login.php

```
<?php
$correctUsername = "admin";
$correctPassword = "1234";
$username = $_POST['username'];
$password = $_POST['password'];
if ($username === $correctUsername && $password === $correctPassword) {
    echo "Login successful! Welcome, $username.";
}
else {
    echo "Invalid username or password.";
}
?>
```

2.Sign up Form with JSON-Based Message Storage using PHP

index.html (front-end)

```
<html>
<head>
    <title>Contact Form</title>
</head>
<body>
    <h2>Contact Us</h2>
    <form action="login.php" method="POST">

        <label>Your Email:</label><br>
        <input type="email" name="email" required><br><br>

        <label>Password:</label><br>
        <input type="password" name="password" required><br><br>

        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

login.php (back-end)

```
<?php
$data = [
    'email' => $_POST['email'],
    'password' => $_POST['password'],
    'password1' => password_hash($_POST['password'], PASSWORD_DEFAULT),
    'time' => date("Y-m-d H:i:s") ];

$file = 'messages.json';
$messages = [];
if (file_exists($file)) {
    $messages = json_decode(file_get_contents($file), true);
}

$messages[] = $data;
file_put_contents($file, json_encode($messages, JSON_PRETTY_PRINT));
echo "Data Stored to JSON file!";
?>
```